Week-1 Presentation

Predicting hourly bike rental demand from weather and seasonal settings

Bike Sharing Dataset | Capital Bikeshare, Washington D.C.



Business Problem

Demand for bikes swings every hour with the time of day and the weather. Guessing wrong costs money either way.

Too few bikes

Customers get frustrated and business is lost.

VS

Too many bikes

Bikes sit idle at stations and resources are wasted.

The challenge: predict how many bikes people will rent at any given hour.

Solution: Predict Demand with ML

Machine learning forecasts bike demand ahead of time, so the business can act before bikes run out.

1

Predict

A Random Forest Regressor forecasts demand from time and weather.

2

Plan

Know when and where bikes will be needed and plan resources.

3

Balance

Keep bikes available across stations so customers always find one.

Result: smooth operations and happy customers.

Data Preprocessing & Cleaning

Getting the data ready before training the model.

17,379

rows, 17 columns

hour.csv holds one record for every hour.

4

columns dropped

casual and registered add up to the target, so they would leak the answer. dteday and instant are only identifiers.

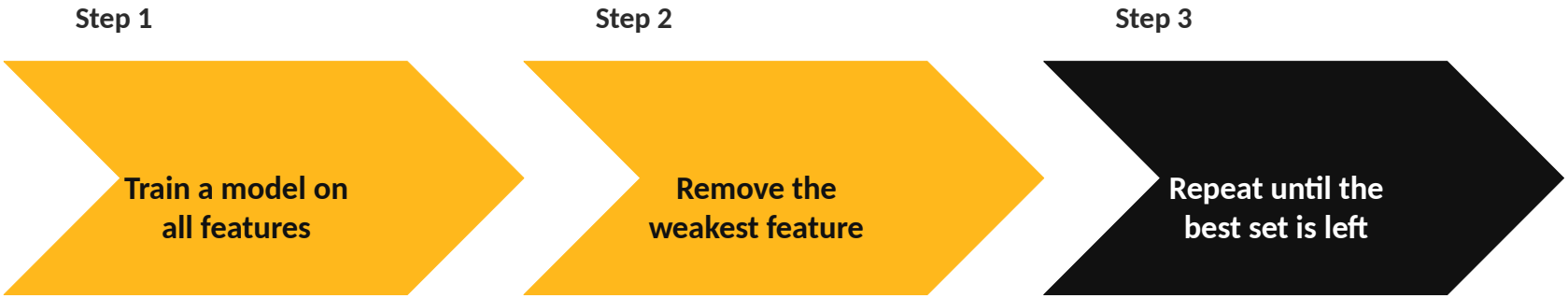
0

missing values

Duplicate rows were removed to keep the data clean and unbiased.

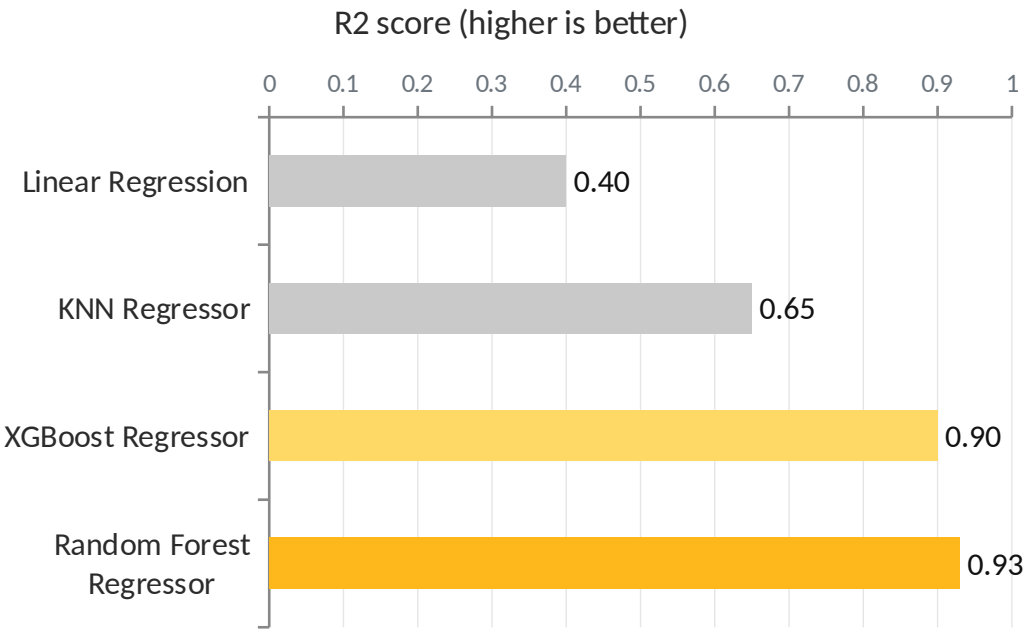
Feature Selection

RFE (Recursive Feature Elimination)



The model tests combinations of features, drops the weakest ones one by one, and keeps only the most important features needed for accurate predictions.

Model Training & Comparison



Approximate values from the Week-1 experiments.

WINNER

Random Forest Regressor

R2 ~0.93

Lowest error (RMSE ~45)

Handles the complex, non-linear patterns in bike rental demand.

Thank You

Questions?

